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Lab 1: A Calculator ISA

Lab 1 is due Monday February 27. Your lab report and source code must be submitted by **10:10AM before the class**. The late policy applies to this lab project.

This lab is a **team project**. All team members are expected to work together on ALL parts of the lab. Get started early! The required format for lab reports can be found on the resource page.

Objective: The objective of this laboratory is to design an 8-bit calculator ISA, design a test benchmark, and do a paper simulation your ISA.

Specification of the calculator: The calculator ISA must support the following operations. You may encode other functions in the ISA as you wish.

- Address four 8-bit registers.
- Load a 4-bit immediate number into a register. The immediate number should be **sign-extended**.
- Add two registers and store the result in a third register.
- Negate a register and store the result in a second register. Here negating means the two's complement of the value in the register.
- Display a register's content to console.
- Skip the next instruction.
- Compare two registers. If $\text{reg1} > \text{reg2}$, execute the next instruction. Otherwise, skip the next instruction.

Task 0: Email the instructor your team composition by Wednesday February 15.

Task 1: Design the ISA. Submit a design document that specifies the format of instructions and the semantics of each instruction. The format should be similar to the MIPS ISA Reference. A sample is provided.

Task 2: Design a benchmark, i.e., a program written **ONLY** in your ISA, to verify the functionality and correctness of your future implementation of the ISA as a BLACKBOX. The benchmark should meet the following requirements:

- (1) Use all operations at least once.
- (2) Each instruction in the benchmark influences the outcome. In other words, whether an instruction is correctly executed in an implementation of your ISA will lead to different outcomes of your benchmark.

Include a binary representation of your benchmark in your submission, e.g., “00110010” in the file. In the lab report, you should explain the design of your benchmark, in particular how it may test the correct implementation of each instruction. Illustrate your benchmark with an on-paper execution and tabulate possible outcomes for successes/failures.

A sample testbench is provided. It should only be used as a format reference. The binary values, as well as the pseudo-code are not relevant.

What to Turn In

For this lab project, turn in your design documents and benchmarks. You should submit all documents to Canvas.

Each submission should include (1) an ISA design document for task 1; (2) the benchmarks you design in binary format, and (3) the documentation of your benchmark design and the explanation of why your benchmark can test an implementation of your ISA.

Peer evaluation: Please submit your peer evaluation (guideline is on Canvas) of your teammates, including scores and justification, directly to the TA.